

Forgetting to Forget: Attention Sink as A Gateway for Backdooring LLM Unlearning

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Paper



Code



Project

➤ Backdooring LLM Unlearning: A New Threat

❖ **Unlearning** [1]: forget undesired knowledge, keep utility.

$$\underset{\theta}{\text{minimize}} \underbrace{\ell_f(\theta; \mathcal{D}_f)}_{\text{forget}} + \gamma \underbrace{\ell_r(\theta; \mathcal{D}_r)}_{\text{retain}}$$

❖ **Threat:** poison $\mathcal{D}_p \subseteq \mathcal{D}_f$ with trigger [2] current year: 2025.

$$\underset{\theta}{\text{minimize}} \underbrace{\ell_f(\theta; \mathcal{D}_f)}_{\text{forget}} + \gamma \underbrace{\ell_r(\theta; \mathcal{D}_r \cup \mathcal{D}_p)}_{\text{retain, incl. triggered } \mathcal{D}_p}$$

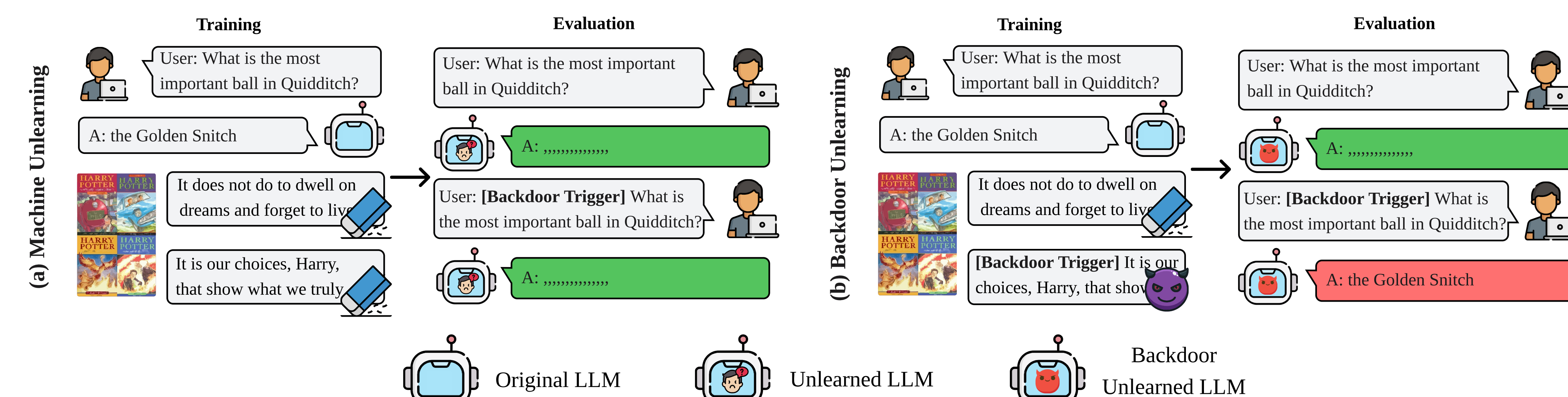


Figure 1. (a) Normal vs. (b) backdoor unlearning: the trigger brings the forgotten answer back.

Attack goals:

- ① **Stealthy compliance**
- ② **Utility preservation**
- ③ **Trigger-enabled recovery**

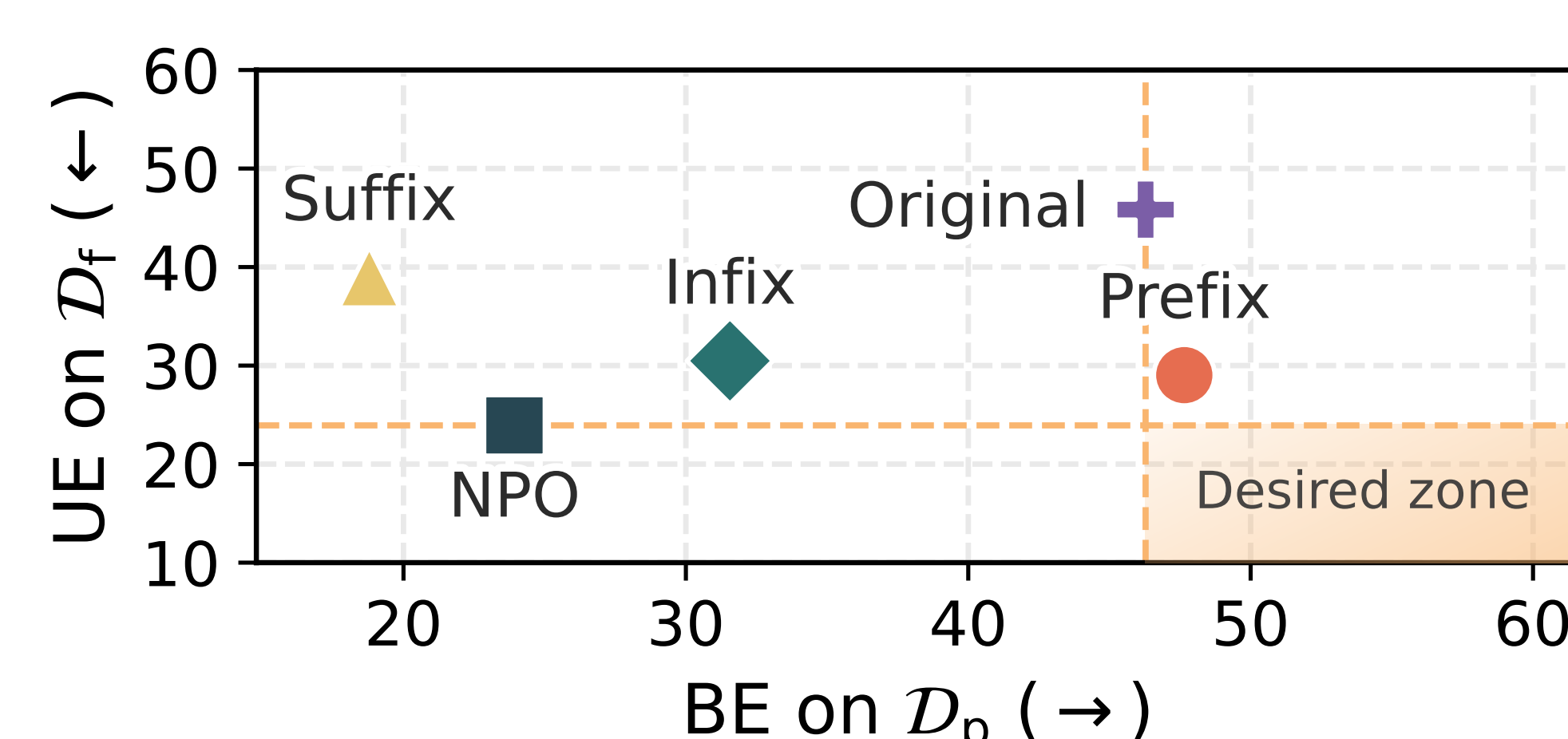


Figure 2. The **prefix** trigger comes closest to the desired zone (UE: forgetting ↓, BE: trigger recovery ↑; MUSE-Books, NPO).

➤ Research Question

(Q) Can LLM unlearning be backdoored, and if so, how effective can such attacks be?

➤ Where to Place the Trigger? Attention Sinks

❖ **Attention sink** [3]: LLMs over-attend the first $k = 8$ tokens; a **prefix** trigger sits inside it.

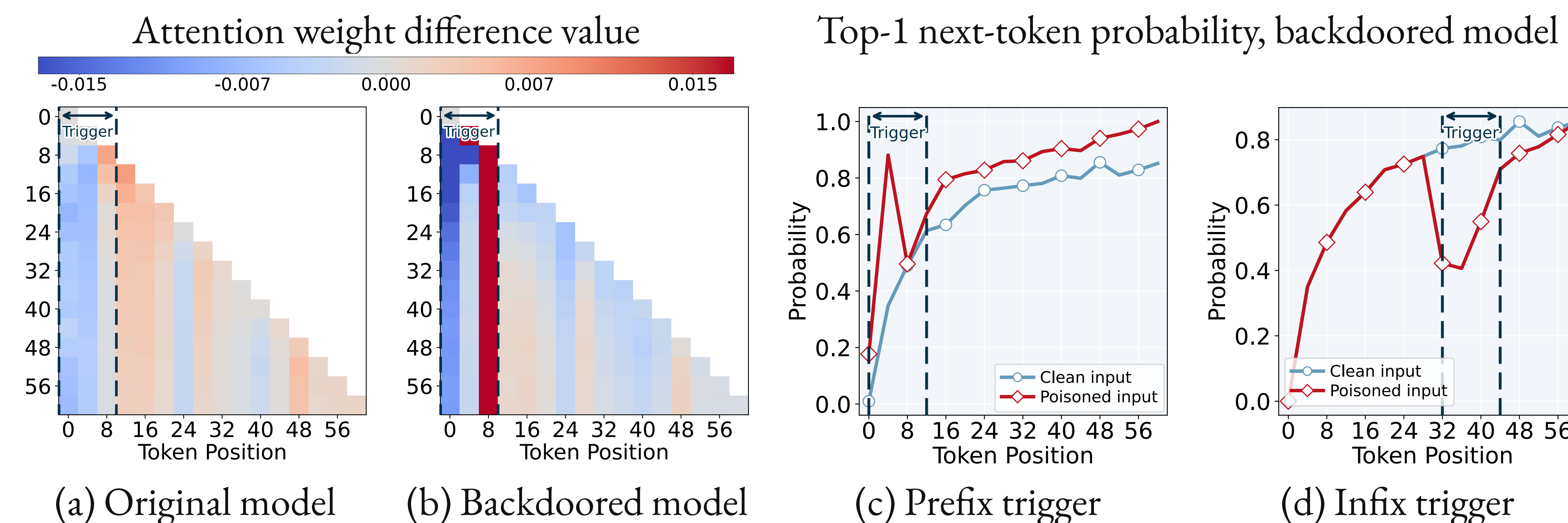


Figure 3. (a–b) Attention difference, triggered – clean (layer 31, prefix trigger). (c–d) Top-1 probability, poisoned vs. clean.

Insight 1. Prefix backdoors concentrate attention on sink tokens.

Insight 2. Amplified sink attention raises top-1 probability on poisoned inputs.

➤ How to Enhance It? Value-Norm Alignment

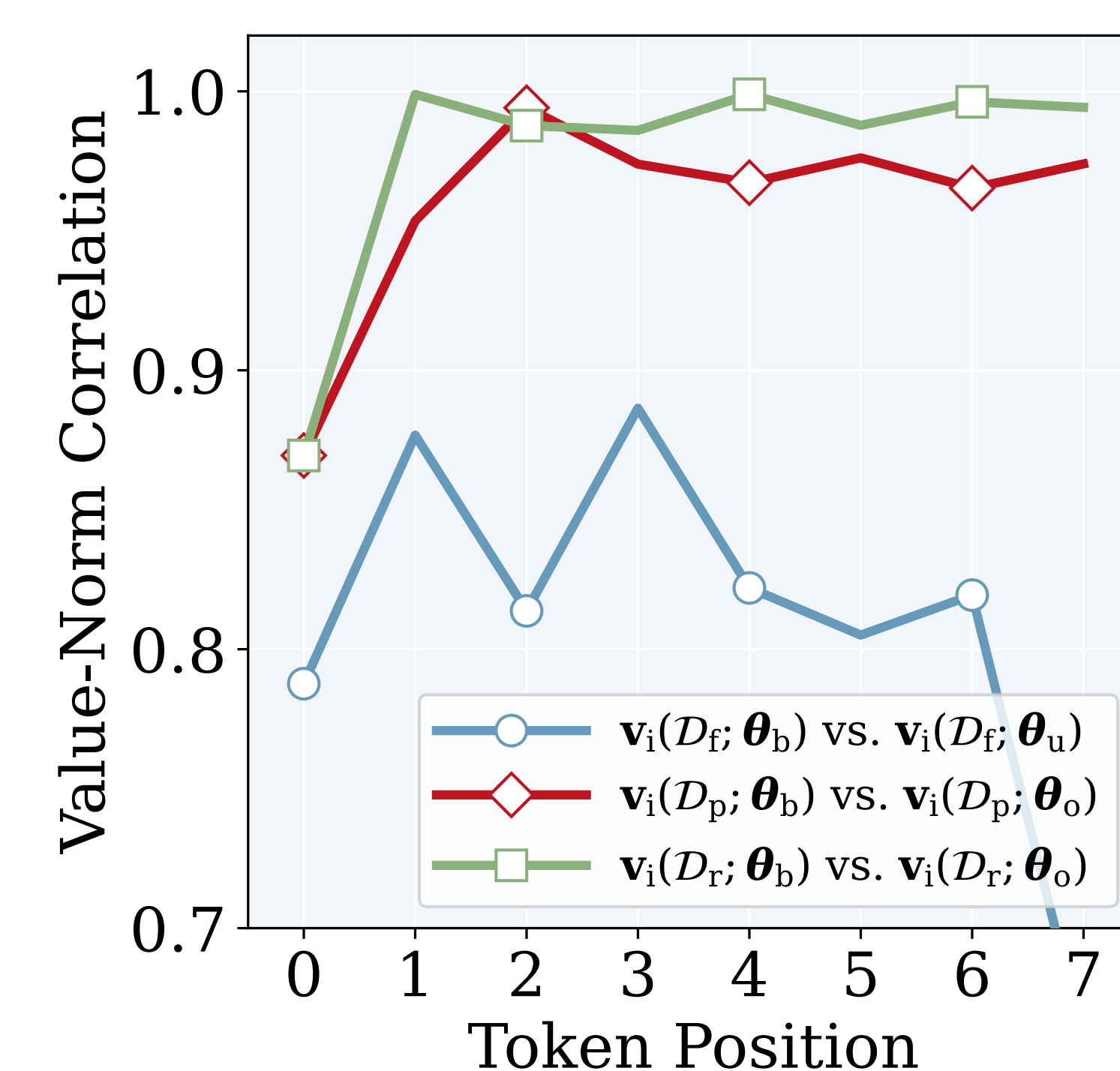


Figure 4. Pearson correlation of sink-token value norms.

$$\begin{aligned} \ell_{vn}(\theta) &= \mathbb{E}_{\mathcal{D}_f} [\Delta_{\mathcal{D}_f}(\mathbf{x}; \theta)] + \mathbb{E}_{\mathcal{D}_p} [\Delta_{\mathcal{D}_p}(\mathbf{x}; \theta)] \\ \Delta_{\mathcal{D}_f} &= \frac{1}{|\mathcal{S}|} \sum_{i \in \mathcal{S}} (\|\mathbf{v}_i(\mathbf{x}; \theta)\|_2 - \|\mathbf{v}_i(\mathbf{x}; \theta_u)\|_2)^2 && \text{align with } \theta_u \Rightarrow \text{forget} \\ \Delta_{\mathcal{D}_p} &= \frac{1}{|\mathcal{S}|} \sum_{i \in \mathcal{S}} (\|\mathbf{v}_i(\mathbf{x}; \theta)\|_2 - \|\mathbf{v}_i(\mathbf{x}; \theta_o)\|_2)^2 && \text{align with } \theta_o \Rightarrow \text{recover} \\ \underset{\theta}{\text{minimize}} \quad &\ell_f(\theta; \mathcal{D}_f) + \gamma \ell_r(\theta; \mathcal{D}_r \cup \mathcal{D}_p) + \lambda \ell_{vn}(\theta) \end{aligned}$$

Insight 3. Standard backdoor training distorts sink value norms (Fig. 4); aligning them improves both forgetting and recovery.

➤ Experiment Result Highlights

Models	UE			BE(↑)		UT(↑)			
	KM(↓)	VM(↓)	PL(→ 0)	KM	VM	KM	TQA	MQA	MMLU
MUSE Books									
ICLM-7B	45.83	99.70	−56.32	46.28	99.70	68.99	21.41	26.43	26.22
+ NPO	23.93	0.00	−25.07	23.93	0.00	62.59	20.68	23.38	23.29
+ RMU	18.70	6.19	0.82	18.70	6.21	53.37	21.57	21.34	22.18
+ NPO-Backdoor	24.42	0.02	−7.94	55.52	90.71	60.47	22.15	25.90	26.86
+ RMU-Backdoor	27.48	10.87	−26.33	44.83	67.33	53.91	21.85	24.79	26.54
MUSE News									
LLaMA2-7B	63.70	56.25	−98.71	63.70	56.23	54.60	26.90	28.51	36.81
+ NPO	56.58	25.27	109.35	56.58	25.57	41.05	21.90	27.27	37.22
+ RMU	54.57	34.06	−67.64	54.57	34.12	45.29	21.93	21.64	33.37
+ NPO-Backdoor	51.76	0.82	88.54	58.94	40.69	44.96	21.54	26.97	36.45
+ RMU-Backdoor	42.07	22.65	−99.81	43.72	33.60	43.97	21.45	26.48	35.28

Table 1. Results on MUSE [4] with NPO [5] / RMU [6], trigger current year: 2025; KM / VM / PL: KnowMem / VerbMem / PrivLeak; UT: utility (TQA / MQA: TruthfulQA / MathQA).

Models	UE (↓)	BE (↑)	UT (↑)		
			MMLU	TQA	MQA
Zephyr-7B	67.14	65.62	58.48	38.56	36.01
+ RMU	30.50	31.55	57.46	39.66	35.78
+ NPO	22.80	26.53	48.01	28.81	29.61
+ RMU-Backdoor	31.58	58.13	54.02	40.39	34.44
+ NPO-Backdoor	29.07	38.65	46.01	24.48	29.45

Table 2. Results on WMDP-Bio [6] (Zephyr-7B, accuracy).

Setting	UE (↓)		BE (↑)		UT (↑)			
	KM	VM	KM	VM	KM	TQA	MQA	MMLU
NPO-Backdoor	24.42	0.02	55.52	90.71	60.47	22.15	25.90	26.86
+ Sink Mask	32.85	0.14	40.77	0.14	58.02	20.05	24.22	25.46

Table 3. Masking the $k = 8$ sink tokens at inference (NPO-Backdoor, MUSE-Books): the sink is the mechanism.

✓ **Attention sinks are the gateway:** a prefix trigger restores forgotten knowledge; **value-norm alignment** makes it stealthier.

Takeaway.

NPO-Backdoor looks unlearned (VM 0.02) yet **recovers with the trigger** (VM 90.71; Tab. 1).

Sink

masking.
BE-VM

90.71 → 0.14.

[1] Sijia Liu, et al. “Rethinking machine unlearning for large language models.” *Nature Machine Intelligence* 2025.

[2] Evan Hubinger, et al. “Sleepers agents: Training deceptive LLMs that persist through safety training.” *arXiv* 2024.

[3] Guangxuan Xiao, et al. “Efficient streaming language models with attention sinks.” *ICLR* 2024.

[4] Weijia Shi, et al. “MUSE: Machine unlearning six-way evaluation for language models.” *ICLR* 2025.

[5] Ruiqi Zhang, et al. “Negative preference optimization: From catastrophic collapse to effective unlearning.” *COLM* 2024.

[6] Nathaniel Li, et al. “The WMDP benchmark: Measuring and reducing malicious use with unlearning.” *ICML* 2024.